Epidemiology of nonsteroidal anti-inflammatory drug-associated gastrointestinal injury.

Nonaspirin, nonsteroidal anti-inflammatory drugs (NSAIDs) are among the most frequently used drugs in many countries. Use of the majority of NSAIDs increases with age, primarily for symptoms associated with osteoarthritis and other chronic musculoskeletal conditions. Population-based studies have shown that, on any given day, 10-20% of elderly people (> or = 65 years old) have a current or recent NSAID prescription. Over a 6-month period in Alberta, Canada, 27% of elderly people were prescribed NSAIDs. Furthermore, in Tennessee (USA), 40% of elderly people received at least one NSAID prescription annually, and 6% had NSAID prescriptions for > 75% of the year. NSAIDs cause a wide variety of side-effects. The most clinically important side-effects are upper gastrointestinal tract dyspepsia, peptic ulceration, hemorrhage, and perforation, leading to death in some patients. Gastrointestinal side-effects are common; the most common NSAID-associated side-effect is epigastric pain/indigestion. Gastrointestinal side-effects are also a frequent reason both for withdrawal of NSAIDs and for co-treatment with another drug. Indeed, in two population-based studies of people aged > or = 65 years, the use of agents to prevent peptic ulcers or relieve dyspepsia was nearly twice as common in regular NSAID users (20-26%) than in non-NSAID users (11%). In Alberta, Canada, it has been estimated that NSAID use accounts for 28% of all prescriptions for anti-ulcer drugs in people aged at least 65 years. Many studies have now shown that NSAIDs increase the risk of peptic ulcer complications by 3-5-fold, and in several different populations it has been estimated that 15-35% of all peptic ulcer complications are due to NSAIDs. In the United States alone, there are an estimated 41,000 hospitalizations and 3,300 deaths each year among the elderly that are associated with NSAIDs. Factors that increase the risk of serious peptic ulcer disease include older age, history of peptic ulcer disease, gastrointestinal hemorrhage, dyspepsia, and/or previous NSAID intolerance, as well as several measures of poor health.